

ROCHESTER INTL, MN, USA

WMO: 726440

Lat: 43.904N

Lon: 92.492W

Elev: 1304

StdP: 14.02

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14925

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-8.1	-18.9	2.1	-12.2	-14.4	2.6	-7.5	31.8	12.9	28.5	15.7	11.2	310	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	87.7	73.5	84.7	71.9	82.3	70.5	76.9	84.2	74.7	81.8	72.9	79.7	13.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
0.4%			1%			2%			0.4%		1%		2%			
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		(p)
74.5	135.5	82.1	72.3	125.5	79.7	70.4	117.5	77.6	41.2	84.2	39.1	81.8	37.3	79.5		85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	25.0	22.6	-19.2	92.7	5.8	2.9	-23.4	94.8	-26.7	96.6	-30.0	98.2	-34.2	100.3
			-19.4	79.9	5.3	2.2	-23.1	81.5	-26.2	82.8	-29.2	84.0	-33.0	85.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	14.8	18.7	31.7	45.6	57.6	67.5	70.8	68.3	61.8	48.6	34.2	20.9	
	DBStd	21.99	13.01	12.44	12.39	9.97	8.67	6.57	5.46	5.31	7.98	9.55	10.92	11.83	
	HDD50	4308	1090	876	580	196	26	0	0	0	8	144	485	903	
	HDD65	7779	1555	1295	1033	585	261	48	13	26	154	516	925	1368	
	CDD50	2548	0	0	13	63	263	526	645	568	360	100	10	0	
	CDD65	545	0	0	1	2	32	124	193	129	57	7	0	0	
	CDH74	4277	0	0	3	42	327	995	1527	867	445	71	0	0	
	CDH80	1084	0	0	0	8	97	275	419	172	98	15	0	0	
Wind	WSAvg	11.4	13.0	12.8	12.6	13.2	12.2	10.4	8.9	8.2	9.6	11.3	12.6	12.4	
Precipitation	PrecAvg	31.6	0.9	0.8	1.9	3.1	3.8	4.4	4.1	4.0	3.5	2.3	1.7	1.1	
	PrecMax	43.9	2.5	2.2	4.0	7.3	12.2	12.5	12.3	14.1	10.5	7.6	5.9	3.7	
	PrecMin	15.4	0.1	0.0	0.3	0.9	1.2	0.9	1.0	0.4	0.4	0.3	0.1	0.2	
	PrecStd	6.4	0.6	0.6	0.9	1.5	2.0	2.4	2.3	2.2	2.5	1.5	1.3	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	49.3	70.6	79.6	88.8	90.9	91.3	88.1	87.3	81.8	67.0	52.4	
		MCWB	38.5	45.0	61.0	62.2	67.8	73.6	77.5	76.5	72.0	65.9	56.0	49.5	
	2%	DB	37.7	42.6	61.0	71.9	82.1	86.6	87.7	84.2	82.7	74.3	59.7	43.8	
		MCWB	34.7	39.7	54.3	56.8	64.5	71.1	75.9	73.4	70.1	61.7	51.8	40.8	
	5%	DB	34.5	38.1	54.5	66.7	77.0	83.0	84.4	81.6	79.2	68.9	54.3	38.9	
		MCWB	32.6	35.5	48.7	53.7	62.7	69.1	73.5	72.4	67.4	59.1	48.2	36.4	
	10%	DB	32.2	35.0	48.0	61.7	72.4	79.7	81.8	78.9	75.5	63.6	49.6	35.5	
		MCWB	30.9	32.9	43.4	51.6	60.2	67.5	71.8	70.2	65.3	55.4	44.3	33.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	45.8	62.0	64.5	71.9	77.0	81.0	78.4	75.1	69.3	58.1	50.4
			MCDB	43.0	48.8	68.3	76.1	84.2	86.7	87.8	85.0	83.1	75.5	64.3	52.7
2%		WB	35.1	40.0	55.5	59.5	67.9	74.1	78.0	75.6	72.2	64.0	53.5	40.9	
		MCDB	37.3	42.4	60.0	68.8	77.2	82.7	85.0	81.8	79.7	70.7	58.7	43.3	
5%		WB	33.0	35.5	49.2	55.8	65.0	71.8	75.5	73.6	69.6	60.2	48.6	36.4	
		MCDB	34.4	37.7	54.5	64.5	73.4	80.1	82.7	79.5	76.6	66.5	53.9	38.6	
10%		WB	30.9	33.1	43.5	52.3	62.2	69.6	73.5	71.7	67.2	56.4	44.6	33.5	
		MCDB	32.3	34.9	47.8	60.3	69.9	77.3	80.2	77.4	73.4	63.3	49.2	35.2	
Mean Daily Temperature Range		5% DB	MDBR	14.4	14.9	15.4	18.8	19.4	18.4	18.1	17.9	20.0	18.2	15.0	13.4
			MCDBR	15.2	14.9	22.6	27.1	25.4	22.1	20.0	19.4	23.0	24.9	22.2	15.8
	MCWBR		13.3	12.6	15.6	15.7	12.6	10.7	10.6	10.9	12.1	13.9	15.3	13.3	
	5% WB	MCDBR	14.1	14.2	21.3	24.1	21.7	19.0	18.3	17.8	20.1	20.3	20.3	14.9	
		MCWBR	12.7	12.3	15.4	15.5	12.4	11.1	11.0	11.0	12.1	13.6	15.2	13.0	
Clear-Sky Solar Irradiance	taub	0.262	0.291	0.314	0.363	0.390	0.400	0.405	0.397	0.366	0.321	0.287	0.261		
	taud	2.415	2.325	2.395	2.314	2.292	2.301	2.287	2.320	2.382	2.494	2.506	2.450		
	Ebn at Noon	279	286	294	285	279	276	273	272	272	273	268	269		
	Edh at Noon	23	30	33	39	41	41	41	38	33	26	22	20		
All-Sky Solar Radiation	RadAvg	556	891	1192	1451	1706	1902	1984	1678	1335	860	562	412		
	RadStd	60	96	118	151	131	144	82	112	95	117	51	37		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page